

Abhinav Singh

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SUMMARY

AI/ML Engineer with 3+ years of experience delivering high-impact machine learning solutions in healthcare and pharmaceutical analytics. Expertise in LLM fine-tuning, statistical modeling (Clustering, Random Forest), and scalable data engineering on Databricks. Proven track record of automating complex workflows and translating business needs into deployable, ROI-driven ML systems. Passionate about Agentic AI, LLM application development, and scalable GenAI systems.

EXPERIENCE

Associate | *Axtria*

May 2024 – Present

- Engineered ML-driven response curve models to replace legacy rule-based systems, increasing targeting precision by **35%** for pharmaceutical outreach.
- Designed and developed an enterprise Retrieval-Augmented Generation (RAG) application using LangChain, embedding models, semantic search, and LLMs to automate business query resolution, reducing response latency by **90%** while improving answer accuracy.
- Architected a scalable data system on **Databricks** for 500K+ physician records, improving processing speed by **50%** and reducing storage costs by **40%**.
- Collaborated with business stakeholders to understand requirements, translate them into AI solutions, and continuously improve application performance based on user feedback.

Analyst | *Axtria*

July 2022 – May 2024

- Automated end-to-end call planning by migrating Excel workflows to SQL, reducing execution time from **8 hours to 45 minutes** with 95% data quality improvement.
- Implemented **Greedy Optimization algorithms** to maximize physician coverage, resulting in a **25% increase** in outreach efficiency.
- Spearheaded cross-functional teams to manage high-stakes client deliverables, ensuring error-free deployment of targeting strategies.

SKILLS

Languages: Python (Scikit-learn, PyTorch), SQL, Java, Pyspark, LangChain, LangGraph

Machine Learning: LLMs (Fine-tuning, RAG), Natural Language Processing, Deep Learning, Statistical Modeling, Optimization Algorithms

Tools & Cloud: Databricks, Apache Spark, Pandas, NumPy, Git, MS Excel

Software Engineering: OOPs, Data Structures & Algorithms, CI/CD Pipelines

PROJECTS

LLM-Based BQC Automated Response System

- Built an enterprise Retrieval-Augmented Generation (RAG) application using LangChain, embedding models, semantic search, and LLMs. Designed an end-to-end pipeline covering document ingestion, chunking, embedding generation, prompt engineering, and response generation. Optimized retrieval accuracy and response latency through context-aware ranking and prompt refinement.

Response Curve Modeling & Segmentation

- Developed machine learning models to optimize physician targeting strategies using response curve modeling and healthcare engagement data. Implemented clustering and predictive modeling techniques to segment healthcare providers based on prescribing behavior and demographic characteristics. Designed scalable feature engineering and model evaluation pipelines to improve targeting precision and marketing ROI. Automated model training and data processing workflows on Databricks, enabling faster execution and reproducible analytics.

EDUCATION

Indian Institute of Information Technology (IIIT)

Allahabad, UP

M.Tech in Software Engineering (GPA: 9.1/10)

2020 – 2022

Madan Mohan Malaviya University of Technology

Gorakhpur, UP

B.Tech in Computer Science Engineering (GPA: 7.1/10)

2016 – 2020